

## ● Original Contribution

### RESULTS OF HIGH DOSE 106-RUTHENIUM IRRADIATION OF CHOROIDAL MELANOMAS

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The favored treatment of intraocular melanomas in Germany is 106-ruthenium eye plaque therapy. The Departments of Radiation Therapy and Ophthalmology (University of Münster) initiated a clinical study in 1981 to reveal the effect of high-dose beta irradiation (15,000 cGy to the apex of the tumor) regarding tumor regression, treatment related side effects, visual acuity, and survival. Sixty-seven patients have been treated since 1981. In 12 patients a second course of irradiation has been performed because of insufficient tumor regression or no change after the first plaque treatment. Sixty-five percent (44/67 pts.) were over 60 years of age. Twenty-four patients had a small tumor (up to 3 mm in height), 20 patients had a medium sized tumor (3.1–5 mm in height), and 22 patients had a large tumor (more than 5.1 mm in height). Fifty-one patients had a follow-up of at least 12 months. A total tumor regression was achieved in 34/51 patients (67%), partial tumor regression occurred in 13/51 patients (25%), and in 4/51 patients (8%) there was no change after the first course. After the second course of 106-ruthenium-irradiation 5 of the 12 patients showed total tumor regression, 3 had partial regression, and in 4 patients only an increase of the tumor echogenicity could be assessed by ultrasonography, but no change in height. Visual acuity, which depends mostly on the localization of the tumor, was preserved at pretreatment levels in 72% of the patients. Two patients died with documented metastatic disease, one patient died of myocardial infarction. There was only one enucleation because of neovascular glaucoma.

**Intraocular melanoma, 106-Ruthenium irradiation, High-dose beta ray therapy, Tumor regression, Visual acuity, Complications**

#### INTRODUCTION

For more than 100 years malignant melanoma of the choroid was the subject of many discussions between ophthalmologists, concerning especially histological classification, diagnostic procedures to prevent diagnostic errors, and optimal treatment strategies. The first detailed presentation on prognosis in ocular melanomas was given by Fuchs<sup>11</sup> in 1882, reporting a mortality rate of 94% in 243 patients, which all were in advanced stages. Because of these observations and the histological examinations of Knapp<sup>20</sup> who described in 1914 an invasion of tumor cells into tumor vessels, exenteration of the orbit was the treatment of choice at that time. Later, Jaensch<sup>18</sup> and von Popolczy<sup>49</sup> showed that prognosis was improved by "early enucleation" of the eye, and the mortality rate was dropped down to 40%; nevertheless, the benefit of early enucleation remained controversial (e.g. von Hippel<sup>48</sup>). Sattler<sup>38</sup> pointed out that  $\frac{2}{3}$  of the patients, regardless of

stage and therapy, died within 3 years. Although we know about the strong influence of histology and phases of growth on prognosis of uveal melanomas,<sup>1,4,36,40,42</sup> in advanced and progressive melanomas, an enucleation of the eye still has to be performed. For early stages the treatment remained controversial, especially since Zimmerman *et al.*<sup>51</sup> stated that enucleation may cause an accelerated dissemination of tumor cells, a problem that was discussed before by Westerveld-Brandon *et al.*<sup>50</sup> and Meyer-Schwickerath.<sup>28,29</sup>

Alternative bulb preserving treatment techniques were developed, including periodic observation.<sup>12</sup> Because of a relatively high rate of side effects, treatment with Radon seeds<sup>30</sup> and also diathermy<sup>21</sup> have not become a routine method in uveal melanomas. Photocoagulation has proven to be successful in small tumors (up to 1 mm of prominence and 3 mm in diameter).<sup>25,28,29,47</sup>

The development of innovative radiation therapy techniques, for example <sup>60</sup>Co plaque therapy,<sup>2,3,10,40-43,45</sup> io-

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**Acknowledgements**—The authors wish to thank G. Held, B. Muhmann and P. Esters for their assistance in preparing the manuscript.

Accepted for publication 28 April 1986.